Multimodal Deep Learning

Exercise Sheet 4 (Week 5)

Date: 27th of May, 2025

Exercise 1: Groups

For each set S and binary operation $*$ below, determine whether $(S, *)$ forms a group. If yes, verify all four group axioms. If no, identify which axiom fails with a counterexample.

(a) $(\mathbb{N}_0, +)$, where $\mathbb{N}_0 = \{0, 1, 2, \dots\}$.

(b) $(R_S, \circ)$, with

$$R_S = \{R_0, R_{90}, R_{180}, R_{270}\}, \qquad R_\theta = \text{rotation of the square by } \theta^\circ \ (0 \le \theta < 360).$$

(c) $(\{R_\theta \mid \theta \in \mathbb{R}\}, \times)$, where

$$R_\theta = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix},$$

and the operation is matrix multiplication.

Exercise 2: Rotational Symmetries in 2D

Let $\mathcal{G} = C_4$ be the group of 90° rotations in the two-dimensional plane, acting on square images of size $H \times W$ (assume $H = W$ for simplicity). Denote by g_{90} the element representing a clockwise rotation by 90°. The action of $g \in C_4$ on an image $\boldsymbol{I}$ is written $g \cdot \boldsymbol{I}$.

(a) Suppose you have a filter (e.g. a Sobel filter) that detects horizontal edges. Let $f(\boldsymbol{I})$ be the feature map obtained by applying this filter to an image $\boldsymbol{I}$. If you apply the filter to a 90°-rotated image, $f(g_{90} \cdot \boldsymbol{I})$, how would you expect this result to relate to $g_{90} \cdot f(\boldsymbol{I})$ (i.e. the original feature map rotated by 90°)? Is this an example of equivariance?

(b) A standard convolutional neural network (CNN) layer is typically composed of a convolution with learned filters, a non-linear activation (e.g. ReLU), and possibly a pooling operation. Is such a CNN layer (with filters learned from data without explicit symmetry constraints) generally equivariant to C_4 rotations? Explain why or why not.

(c) You are designing a neural network to analyse cell-microscopy images in which cell orientation is arbitrary and diagnostically irrelevant. Why would C_4 (or continuous $SO(2)$) equivariance or invariance be a desirable property for your network? Which parts of the network would you aim to make equivariant, and which parts invariant?

Exercise 3: Equivariance and Invariance

(a) Consider the task of object detection in images. An image $\boldsymbol{x} \in \mathbb{R}^{H \times W \times C}$ is input, and the output is a set of bounding boxes. If the image is translated, how should the bounding boxes ideally transform? Does this describe equivariance or invariance with respect to the translation group? Explain.

(b) Consider the task of image classification. An image $\boldsymbol{x} \in \mathbb{R}^{H \times W \times C}$ is input, and the output is a class label (e.g., "cat" or "dog"). If the image is slightly rotated, should the class label change? Does this ideally describe equivariance or invariance with respect to a rotation group?

Exercise 4: Coding Exercise: Rotation Equivariance and Group Convolutions

Solve the exercises in the accompanying file "exercise4.ipynb".

Exercise 5: Applications

You are designing a CNN for detecting lung nodules in chest X-rays. The nodules can appear at any orientation.

(a) Explain why rotation equivariance would be beneficial for this task.

(b) Compare three approaches:

- Standard CNN with data augmentation
- Group-equivariant CNN with C_4 symmetry
- Steerable CNN with $SO(2)$ symmetry

Discuss the trade-offs in terms of:

- Parameter efficiency
- Computational cost
- Expected performance

(c) Which approach would you recommend and why? Consider that medical datasets are typically small.

Exercise 6: Translation Equivariance

Let T_a denote translation by vector $\mathbf{a} \in \mathbb{R}^2$, and let $*$ denote the convolution operation. Assume zero-padding at the boundaries.

(a) Prove that standard 2D convolution is translation-equivariant:

$$\mathrm{T}_{\mathbf{a}}(f * k) = (\mathrm{T}_{\mathbf{a}} f) * k$$

(b) Given a 1D signal $x = [1, 2, 3, 4]$ and kernel $k = [1, -1]$, compute:

The convolution $y = x * k$
The translated signal $x' = T_1(x)$ (shift by 1)
Verify that $T_1(y) = x' * k$